ANIKET'S SMART HOME ENERGY MONITOR

A Python-Based Simulation for Household Electricity Monitoring and Energy Awareness

PROJECT REPORT

Extracurricular Activity (ECA) • Engineering / Programming

Platform: Python and Tkinter **Operation:** Local desktop simulation

Core focus: Energy calculation, appliance management, data presentation and recommendations

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Note: This project uses simulated readings and user-entered appliance data. It does not connect to physical sensors or a live smart meter.

Contents. Report Overview

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This report describes the project's purpose, design, calculations, features, testing approach and possible future development. The values shown as examples are illustrative and should be replaced with verified screenshots or measured outputs if required by the school.

1. Abstract

Aniket's Smart Home Energy Monitor is a desktop application developed in Python to demonstrate how household electricity use can be estimated, organised and presented through a graphical interface. The project was created as an independent engineering- and programming-focused extracurricular activity. It allows a user to manage a list of household appliances, enter their power ratings and typical operating hours, estimate energy consumption and electricity cost, and review the results through dashboard-style pages. The interface is built with Tkinter, Python's standard graphical user-interface toolkit, and is designed to run locally on a computer.

The application also explores the presentation of simulated electrical readings, energy insights, city-average recording and practical energy-saving suggestions. These elements provide a foundation for discussing how a future smart-home system could help users understand consumption patterns. The current version is a software simulation: it does not measure real voltage, current or power, and its outputs depend on entered data and programmed assumptions.

Item	Description
Project type	Independent software simulation / ECA
Programming language	Python
User interface	Tkinter desktop GUI
Primary purpose	Estimate household energy use and communicate energy-saving insights
Data source	User-entered appliance settings and simulated readings

2. Introduction and Motivation

Electricity is used throughout a home for lighting, cooling, entertainment, computing and other daily activities. However, the energy impact of individual appliances is not always obvious. A device with a high power rating may consume substantial energy even when used for a relatively short time, while smaller devices can add up when they operate for many hours. A simple monitoring interface can make these relationships easier to understand.

This project applies basic programming and electrical-energy concepts to a practical household scenario. Instead of connecting to a physical meter, the application models appliances using their power ratings and estimated daily usage. This approach makes it possible to explore calculations, user-interface design, data handling and energy-awareness features without additional hardware.

The project is relevant to electrical engineering because it uses electrical quantities such as power, voltage and energy as the basis for a software tool. It is also relevant to computer science because the program involves a graphical interface, application logic, data organisation and report-style outputs.

3. Project Objectives

• Build a clear desktop interface for viewing household energy estimates.

• Allow appliance details, power ratings and usage hours to be edited.

• Calculate estimated daily and monthly energy consumption in kilowatt-hours (kWh).

• Estimate electricity cost using a configurable tariff.

• Present key figures and insights in separate sections rather than a single screen.

• Demonstrate simulated live readings and a city-average recording concept.

• Offer understandable suggestions that encourage more efficient electricity use.

• Provide a foundation for future integration with real sensors or smart-meter data.

Success criteria

The project is considered to meet its basic goals when the application opens successfully, appliance settings can be edited, calculations respond to changes in inputs, the main pages can be accessed, and exported reports or data can be saved where those functions are implemented. The program should also clearly identify simulated values so they are not mistaken for real measurements.

4. System Design and Technology

The application can be understood as a small local software system with four main layers:

Layer	Role in the project
Input layer	Appliance names, rated power in watts, usage hours, electricity tariff and other user settings.
Calculation layer	Converts watts to kilowatts and combines power with time to estimate energy and cost.
Presentation layer	Tkinter windows, navigation tabs, summary cards, tables, charts or text-based insights.
Output layer	Displayed estimates, recommendations and exported data or reports where enabled.

Technology choices

Technology	Purpose
Python	Core language for application logic and calculations.
Tkinter / ttk	Desktop interface, controls, tables and page layout.
CSV / local files	A lightweight way to save or export structured data when used by the application.
Mathematical formulas	Energy and cost estimation from appliance power and operating time.

The application is intended to run locally and does not require a web server. Exact library use and export formats should be checked against the final source-code version before submission.

5. Main Features

Feature	Purpose and user value
Overview dashboard	Shows a quick summary of estimated energy use, cost and key household figures.
Appliance management	Lets the user review or edit appliance power ratings and expected daily operating hours.
Energy insights	Organises calculated results to help identify appliances with comparatively high estimated energy use.
Simulated live readings	Displays changing example readings to demonstrate how a live-monitoring panel might behave; not sensor data.
City-average recording	Provides a concept for recording and comparing average energy figures associated with cities or locations. Its accuracy depends on the source and method used for the entered data.
Suggestions	Presents practical ideas such as reducing unnecessary operating time and checking high-consumption appliances.
Reports and export	Supports saving or exporting information where implemented, making results easier to review outside the application.

Typical appliance examples

Appliance	Illustrative rated power	Example daily use
LED bulb	10 W	6 hours
Ceiling fan	100 W	8 hours
Laptop	65 W	5 hours
Television	100 W	4 hours
Refrigerator	150 W	8 hours equivalent run time
Water pump	750 W	1 hour

These are example settings previously used during project development, not laboratory measurements. Real appliance consumption varies by model, load, operating mode and actual runtime.

6. Energy Calculation Method

The application uses standard energy-estimation relationships. Electrical energy is calculated from power multiplied by the time for which the appliance operates.

Daily energy consumption

Energy (kWh) = Power (W) ÷ 1,000 × Operating time (hours)

Estimated monthly energy

Monthly energy (kWh) = Daily energy (kWh) × Number of days

Estimated electricity cost

Estimated cost = Energy (kWh) × Tariff per kWh

For example, a 100 W fan used for 8 hours per day has an estimated daily energy use of 0.1 kW × 8 h = 0.8 kWh. Over 30 days, the estimate is 24 kWh. At an illustrative tariff of NPR 10 per kWh, the corresponding estimated cost is NPR 240. This example explains the calculation and is not a statement of the user's actual electricity bill or the official current tariff.

Example input	Value
Rated power	100 W
Daily use	8 hours
Daily energy	0.8 kWh
Monthly energy (30 days)	24 kWh
Illustrative tariff	NPR 10/kWh
Illustrative monthly cost	NPR 240

Calculation assumptions

The estimate assumes the appliance operates at its entered rated power for the stated duration. Actual devices may draw variable power, and refrigerators or pumps may cycle on and off. Electricity bills can also include slab-based pricing, fixed charges, taxes or other adjustments not represented by a single flat tariff.

7. Implementation and Workflow

Step	What happens
1. Launch	The user opens the desktop application.
2. Enter or edit data	Appliance details and usage assumptions are reviewed or changed.
3. Calculate	The program uses power and time values to estimate kWh and cost.
4. Review	The dashboard and insight pages present the calculated results.
5. Explore suggestions	The user reviews possible ways to reduce unnecessary energy use.
6. Save or export	Where supported, the user exports records or a report for later review.

Development approach

The project was developed iteratively. A basic energy-monitoring concept was expanded into a more complete desktop application with multiple sections, editable appliance information and reporting-oriented features. During development, attention was given to making the interface easier to navigate and to separating user-facing information from the underlying calculations. A useful next step is to test every button and exported file in the exact final version and capture screenshots for evidence.

8. Testing and Observations

Testing for a software simulation should verify both the arithmetic and the behaviour of the interface. The table below is a recommended test record. Mark each test as passed only after running it in the final program; the report does not claim that unverified tests have already passed.

Test	Procedure	Expected result
Application launch	Run the Python file.	Main window opens without a fatal error.
Appliance edit	Change a power rating or usage time.	The edited value is reflected in the relevant calculations.
Energy formula	Use 100 W for 8 hours.	Daily estimate equals 0.8 kWh.
Monthly estimate	Set 30 days for the same example.	Monthly estimate equals 24 kWh.
Tariff update	Change tariff to NPR 10/kWh.	The example monthly cost equals NPR 240.
Input validation	Try a negative power value or invalid duration.	The program rejects it or handles it clearly, if validation is implemented.
Export function	Use the available save/export control.	A file is created and can be opened, if export is implemented.
Simulation labelling	Review live-reading panel.	Simulated data is not presented as real sensor measurements.

Observations

The project demonstrates that a desktop program can turn appliance assumptions into estimates that are easier to interpret than a list of wattages alone. It also highlights an important engineering distinction: a visual display of changing values is not the same as real-time measurement. Reliable monitoring would require suitable sensors or a meter, calibration, safe electrical interfacing and a verified method of acquiring data.

9. Limitations

• The current system is a simulation and does not connect to physical current, voltage or power sensors.

• Estimated consumption depends on the accuracy of the entered wattage and usage hours.

• Changing simulated readings are illustrative and must not be used for electrical safety decisions.

• City-average comparisons are only meaningful if their source, location, time period and measurement method are documented.

• A single flat tariff does not capture every detail of a real electricity bill.

• The quality of charts, exports, input validation and persistence depends on what is implemented in the final source-code version.

These limitations do not remove the project's educational value. They define its scope honestly and identify what would be needed to progress from a software demonstration to a practical monitoring system.

10. Future Improvements

Improvement	Possible development
Physical measurement	Connect a properly rated energy-monitoring sensor or certified smart meter using a safe, isolated interface.
Persistent history	Store daily readings in a database or structured file so users can compare weeks and months.
More accurate billing	Add tariff slabs, service charges and other relevant billing rules.
Validated city data	Use a clearly cited and consistently defined dataset rather than unsupported city averages.
Alerts and targets	Allow users to set monthly kWh or cost targets and show warnings when estimates exceed them.
Better evaluation	Compare the software's estimates with readings from a real meter and report the difference.
Packaging	Package the application as a Windows executable after testing dependencies and file paths.

Safety consideration

Any future hardware version involving mains electricity must use appropriately rated, enclosed equipment and safe isolation. Directly wiring a homemade circuit to household mains is not an appropriate beginner experiment. A certified plug-in energy meter or professionally designed isolated sensor arrangement is safer for validation.

11. Learning Outcomes

• Applied the relationship between electrical power, operating time and energy consumption.

• Practised structuring a Python program into calculations, data inputs and interface elements.

• Explored desktop GUI design, navigation and the presentation of summary information.

• Considered how data export and historical records can support analysis.

• Learned to distinguish simulated data from measurements and to state assumptions clearly.

• Identified realistic future work, including validation, data quality and hardware integration.

The most important engineering lesson is that a useful monitoring interface depends not only on appearance but also on transparent assumptions, correct calculations and trustworthy data. The project provided a practical way to explore these principles using accessible software tools.

12. Conclusion

Aniket's Smart Home Energy Monitor is a student-developed software simulation that brings together basic electrical-energy calculations and desktop application design. Its appliance-management, dashboard, insight, simulated-reading and recommendation concepts demonstrate how software can help communicate household energy use. The project is suitable as an ECA because it shows independent exploration of programming and an engineering-related problem while remaining feasible without specialised hardware.

The next stage is to validate the final application carefully, capture screenshots of the working interface, verify each calculation and document the source of any comparison data. If developed further, the project could incorporate a trusted meter, persistent history and more realistic billing logic. Until then, its results should be understood as estimates based on user-entered inputs rather than actual measurements.

Appendix. Sample Calculation and Project Summary

Worked example: ceiling fan

Parameter	Calculation	Result
Power in kilowatts	100 W ÷ 1,000	0.1 kW
Daily energy	0.1 kW × 8 h	0.8 kWh
Monthly energy	0.8 kWh × 30 days	24 kWh
Monthly cost	24 kWh × NPR 10/kWh	NPR 240

Project summary

Title: Aniket's Smart Home Energy Monitor
Purpose: Estimate and present household electricity consumption for learning and energy awareness.
Tools: Python, Tkinter and local data/export functions where implemented.
Scope: Software simulation using editable appliance assumptions and simulated readings.
Evidence to attach: Screenshots of the overview, appliance editor, city-average tab, suggestions page and a sample exported report, taken from the final working version.

Prepared as an extracurricular project report. Update the date, add verified screenshots, and confirm that each feature described matches the exact source code being submitted.